BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MARDAN

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2023

CHEMISTRY

Past Paper 2023 • Paper Code: MRD-9th 130525 (A)

Retyped faithfully from the original board paper — same questions, no changes.

Section – A

Time Allowed: 20 Minutes | **Maximum Marks:** 12 | **Paper Code:** MRD-9th 130525 (A), VERSION: A

Note: There are three sections in this paper i.e A, B & C. Instructions: Please attempt Section-A on the MCQs Answer Sheet only. Only black ball point or marker may be used for shading the relevant circle. Please make sure that there is no cutting, over writing, erasing, or multiple circles shading.

Q1. Choose the correct option i.e A, B, C or D.

Q1. Which one of the following has sharp and fixed melting point.

(A) Homogenous mixture
(B) Compound
(C) Heterogonous mixture
(D) Impure substance

Q2. Why oxide ion has -2 charge.

(A) Oxygen looses two electrons
(B) Oxygen gains two electrons
(C) Oxygen becomes neutral
(D) Oxygen looses three electrons

Q3. The number of particles approximately emitted from polonium metal are:

(A) 28,000
(B) 24,000
(C) 22,000
(D) 20,000

Q4. The rarest elements in the earth crust is:

(A) Astatine
(B) Iodine
(C) Chlorine
(D) Bromine

Q5. Sodium chloride (NaCl) and Calcium Chloride ($CaCl_2$) are the examples of:

(A) Electrovalent Bond
(B) Covalent Bond
(C) Metallic Bond
(D) Dative Bond

Q6. Which one of the following pairs of atoms is most likely to form a covalent bond:

(A) Na and F
(B) K and F
(C) Ca and F
(D) F and F

Q7. Which process occurs when we spray a perfume in corner of room?

(A) Effusion
(B) Diffusion
(C) Osmosis
(D) Ellusion

Q8. How many grams are required to prepare one Molar solution of NaCl?

(A) 54gm
(B) 56gm
(C) 58gm
(D) 60gm

Q9. A mixture of starch in water is an example of:

(A) Colloids
(B) Solution
(C) Suspension
(D) Solvant

Q10. In NaCl, the element Na is:

(A) Or:idizing Agent
(B) Reducing Agent
(C) Polarizing Agent
(D) Decolorizing Agent

Q11. Which of the following compound shows minimum oxidation number of carbon?

(A) H_2CO^3
(B) CO_3
(C) CO_2
(D) CO

Q12. Which one of the following element is metalloid?

(A) Lithium
(B) Beryllium
(C) Nitrogen
(D) Boron

Section – B

Time Allowed: 2.40 Hours | **Maximum Marks:** 32

Q2. Write short answers to any eight (8) of the following parts. All parts carry equal marks:-

(i) How many grams are present in 4 moles of NaOH? (At. mass Na=23amu, O=16amu, H=1amu)

(ii) Write difference between Rutherford and Bohr's atomic model?

(iii) How many shells and valence electrons, the elements Chlorine and Magnesium have?

(iv) Explain the Octet rule with the help of electronic configuration of inert gases?

(v) Give difference between single and triple covalent bond with the help of example?

(vi) A 500ml sample of hydrogen is heated from 27°C to 77°C at constant pressure. What is the final volume of the gas?

(vii) Give one example of the following solutions: (a) Gas in Gas (b) Gas in Liquid (c) Liquid in Gas (c) Liquid in Solid

(viii) How can we prepare 500ml of 0.02 molar Solution from 1 molar solution of HCl?

(ix) Write chemical reactions for manufacturing of NaOH from Brine.

(x) Explain electroplating of Zinc in terms of cathodic and anodic reaction.

(xi) Why gold is used in standard desktop or laptop computers. Give proper reasons?

Section – C

Maximum Marks: 21

> Note: Answer any three (3) questions. All questions carry equal marks:-

Q3.

(i) [3] Find out the molecular mass of C_2H_6 in a.m.u and also expressed it into grams?

(ii) [4] Write the names of isotopes of Chlorine. Draw their atomic structure.

Q4.

(i) [3] Which group in periodic table represents halogens? Write the names of elements included in that group.

(ii) [4] Why are metallic bonds so strong? Justify your answer.

Q5.

(i) [3] Calculate the final volume of $3.5dm^3$ gas at 300mm of Hg when pressure changes to 400m of Hg.

(ii) [4] Differentiate between solute and solvent with the help of an example.

Q6.

(i) [3] Write reaction of sodium with Sulphur and halogens?

(ii) [4] What is the composition of electrodes and which electrolytes are used in dry cell?